ASTRONOMY OLYMPIAD 2026

JUNIOR + SENIOR TARGETED STUDENT SYLLABUSES

Study topics mapped to both practice papers

These are workshop preparation syllabuses, not official organiser-issued syllabuses. They list the concepts and skills students need for the matching model papers.

How to use the pack

Junior students should follow the Junior section only. Senior students should follow the Senior section. Each unit includes the question numbers it supports, exact topics to study, and expected learning outcomes.

JUNIOR CATEGORY - TARGETED SYLLABUS

Mapped directly to the matching practice paper

Expected level: clear concepts, familiar examples, and simple reasoning.

Unit 1. The Solar System and its objects

Practice-paper coverage: MCQ 1, 4, 5, 17

• The Sun is the central star of our Solar System; planets orbit the Sun.

• The eight planets in order from the Sun: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune.

• Jupiter is the largest planet in the Solar System.

• Saturn is especially well known for its prominent ring system.

• The four inner planets are Mercury, Venus, Earth and Mars.

By the end of this unit, students should be able to: Students should be able to name the Sun as the centre of the Solar System, identify Jupiter and Saturn from these facts, and list the four inner planets in order.

Unit 2. Earth, the Moon and the Sun

Practice-paper coverage: MCQ 2, 3, 6, 12; short answer 16

• Earth’s rotation on its axis causes day and night.

• Earth’s axis is tilted. As Earth orbits the Sun, this tilt changes sunlight angle and day length and causes seasons; seasons are not caused mainly by Earth being closer to or farther from the Sun.

• The Moon does not usually make its own visible light; it reflects sunlight.

• Moon phases occur because, as the Moon orbits Earth, observers see different portions of the Moon’s sunlit half.

• A lunar eclipse occurs when Earth lies between the Sun and Moon so that Earth’s shadow falls on the Moon.

By the end of this unit, students should be able to: Students should be able to explain day/night, seasons and phases in simple language, and distinguish a lunar eclipse from a solar eclipse.

Unit 3. Stars, planets and light

Practice-paper coverage: MCQ 7, 8; short answer 18

• A star is a hot ball of gas/plasma that produces energy; the Sun is the nearest star to Earth.

• Stars produce their energy mainly through nuclear fusion. For this paper, understand the idea only: hydrogen nuclei combine to form helium and release energy.

• A star emits its own light. A planet is seen mainly by reflected starlight and does not shine like a star by sustained core fusion.

• Stars appear to twinkle because turbulence in Earth’s atmosphere bends and disturbs incoming starlight. This is atmospheric refraction/scintillation at an introductory level.

By the end of this unit, students should be able to: Students should give one accurate difference between a star and a planet, and describe why stars seem to twinkle.

Unit 4. Comets, satellites and telescopes

Practice-paper coverage: MCQ 9, 10, 14; short answer 19

• Comets contain ice, dust and rocky material. When a comet approaches the Sun, heating releases gas and dust and a visible tail may form.

• A satellite is an object that orbits a larger body. Satellites can be natural (such as the Moon) or artificial (made and launched by people).

• Uses of artificial satellites include communication, weather forecasting, navigation, Earth observation and scientific research.

• A telescope collects electromagnetic radiation/light so distant astronomical objects can be observed more effectively. Different kinds of telescopes detect different wavelength ranges; detailed telescope engineering is not required.

By the end of this unit, students should be able to: Students should describe a comet, define an orbiting satellite, name one practical satellite use, and state the basic purpose of a telescope.

Unit 5. Our galaxy and planets beyond the Solar System

Practice-paper coverage: MCQ 13, 15

• The Milky Way is the galaxy that contains our Solar System.

• An exoplanet is a planet outside our Solar System, usually orbiting another star.

• Know the difference between the Solar System and a galaxy: the Solar System is the Sun and the bodies bound to it; a galaxy contains many stars and other material.

By the end of this unit, students should be able to: Students should identify our galaxy and define an exoplanet.

Unit 6. Simple numerical reasoning: light intensity

Practice-paper coverage: short answer 20

• Learn the inverse-square idea: for light spreading out from the same source, intensity decreases with the square of distance.

• Formula: intensity is proportional to $1/d^2$, where d is distance from the same source.

• If distance becomes 2 times larger, intensity becomes 1/4. If distance becomes 3 times larger, intensity becomes 1/9.

• Practise simple ratios and squares; a calculator should not be necessary for the example in the paper.

By the end of this unit, students should be able to: Students should explain that the more distant planet receives less light and calculate the 1/9 comparison when distance is tripled.

SENIOR CATEGORY - TARGETED SYLLABUS

Mapped directly to the matching practice paper

Expected level: conceptual understanding, formula substitution, simple ratios, and basic unit awareness.

Unit 1. Gravity, orbits and Kepler’s laws

Practice-paper coverage: MCQ 1, 2; short answer 16

• Gravity attracts planets toward the Sun while their forward motion keeps them moving around it; an orbit is continuous motion under gravity, not a planet being held up by air or magnetism.

• Kepler’s third law for objects orbiting the Sun: $T^2 = a^3$ when T is measured in Earth years and a in astronomical units (AU).

• Know what T means (orbital period) and a means (average orbital distance/semi-major axis).

• Rearrange the relationship simply: a = cube root of T^2. For T = 2 years, $a^3 = 4$ and a is approximately 1.59 AU.

• No derivation of Kepler’s laws or advanced orbital mechanics is required for this paper.

By the end of this unit, students should be able to: Students should state Kepler’s third law with the correct units, use it for the given example, and explain the role of gravity in orbits.

Unit 2. Stellar light: luminosity, flux and distance

Practice-paper coverage: MCQ 4, 5, 13; short answer 17

• Luminosity (L) is the total energy a star emits per unit time. It is an intrinsic property of the star.

• Observed flux/brightness (F) is the energy received per unit area at the observer. For the same type of comparison, F is proportional to L/d^2.

• Use ratios: $F■/F■ = (L■/L■) \times (d■/d■)^2$. If luminosity is 4 times greater and distance is 2 times greater, flux changes by $4/2^2 = 1$, so the observed flux is equal.

• Stellar parallax is the apparent shift of a nearby star against distant background objects as Earth changes position in its orbit.

• Distance in parsecs: d(pc) = 1/p(arcseconds). For p = 0.5 arcseconds, d = 2 parsecs.

• Know the terms apparent brightness, luminosity, distance, parallax, parsec and AU.

By the end of this unit, students should be able to: Students should distinguish luminosity from observed brightness, calculate a simple flux ratio, and use the parallax-distance formula.

Unit 3. Temperature and colour of stars

Practice-paper coverage: MCQ 3

• Objects emit a range of wavelengths; a hotter star has its peak emission at a shorter wavelength.

• This is the qualitative idea behind Wien’s displacement law. For this practice paper, no numerical use of Wien’s law is required.

• Know the trend: hotter - shorter peak wavelength; cooler - longer peak wavelength.

By the end of this unit, students should be able to: Students should correctly predict how the peak wavelength changes as surface temperature increases.

Unit 4. The Sun, fusion and stellar lifetimes

Practice-paper coverage: MCQ 6, 7

• The Sun is powered mainly by nuclear fusion in its core: hydrogen nuclei combine through a chain of reactions to form helium and release energy.

• Massive stars generally have higher luminosities and consume their nuclear fuel faster than lower-mass stars.

• As a result, massive stars generally have shorter lifetimes even though they begin with more material/fuel.

• For this paper, understand the broad relationship; detailed pp-chain branches, CNO cycle calculations and advanced stellar structure are not required.

By the end of this unit, students should be able to: Students should identify fusion as the Sun's energy source and explain why massive stars generally live for less time.

Unit 5. Redshift and the expanding universe

Practice-paper coverage: MCQ 8, 9; short answer 20

• Redshift means observed spectral features are shifted to longer wavelengths than their rest wavelengths.

• Redshift can be described by z = (observed wavelength - rest wavelength) / rest wavelength. The redshift z is dimensionless.

• Distant galaxies commonly show redshift, and the observed relation between galaxy distance and recession speed is key evidence that the universe is expanding.

• For the simple low-redshift approximation in the practice paper: $v \approx cz$, where $c \approx 300{,}000$ km/s. For $z = 0.1$, $v \approx 30{,}000$ km/s.

• This is an introductory approximation only; students do not need advanced cosmology or relativistic redshift calculations.

By the end of this unit, students should be able to: Students should identify redshift, state its basic meaning, describe its connection to cosmic expansion, and calculate v using the supplied approximation.

Unit 6. Exoplanets and detection by transit

Practice-paper coverage: MCQ 10

• An exoplanet orbits a star beyond our Solar System.

• Transit detection: when a planet passes in front of its star from our viewpoint, it blocks a small fraction of starlight, causing a repeated dip in measured brightness.

• Understand the light-curve idea at a conceptual level; no detailed orbital geometry or data-fitting is required.

By the end of this unit, students should be able to: Students should identify the transit method from a description of a repeated dip in a star's brightness.

Unit 7. Telescopes and electromagnetic wavelengths

Practice-paper coverage: MCQ 11; short answer 18

• A telescope collects electromagnetic radiation from astronomical objects. Radio telescopes detect radio wavelengths and reveal sources that emit radio waves.

• Infrared light has longer wavelengths than visible light. Some infrared wavelengths pass through certain dust clouds more effectively than visible light, so infrared observations can reveal otherwise obscured regions and cooler objects.

• Not all dust is transparent to infrared light; the key is that infrared can penetrate some dusty regions better than visible light.

• Know the basic distinction between radio and infrared observations; advanced instrument design is outside this paper's scope.

By the end of this unit, students should be able to: Students should match radio telescopes with radio waves and explain why infrared observations can be useful in dusty regions.

Unit 8. Compact stellar remnants and black holes

Practice-paper coverage: MCQ 12; short answer 19

• A black hole is an object whose gravity is so strong that beyond its event horizon, light cannot escape to a distant observer. The event horizon is a boundary, not a solid surface.

• A white dwarf is a compact remnant left by a low- or intermediate-mass star after its outer layers are shed.

• A neutron star is a much denser remnant that can form from the collapsed core of a massive star after a supernova.

• For this paper, compare the broad origin and nature of white dwarfs and neutron stars; no equations for black holes, general relativity or neutron-star structure are required.

By the end of this unit, students should be able to: Students should define an event horizon in simple terms and state one clear difference between a white dwarf and a neutron star.

Unit 9. Dark matter: evidence from gravity

Practice-paper coverage: MCQ 14

• Dark matter is inferred because the gravitational effects seen in galaxies and other systems cannot be explained by the visible matter alone in the simplest account.

• For this paper, know the idea that dark matter contributes gravitational influence while not emitting or reflecting enough light to be directly seen in the usual way.

• Detailed particle candidates, mathematical galaxy models and advanced cosmology are not required.

By the end of this unit, students should be able to: Students should identify dark matter as an inference from gravitational effects, not as ordinary bright material.

Unit 10. Earth’s axial tilt and seasons

Practice-paper coverage: MCQ 15

• Earth’s axis is tilted relative to its orbital plane.

• As Earth orbits the Sun, the tilt changes the angle of incoming sunlight and the length of daylight in each hemisphere.

• These differences produce the seasons; the two hemispheres experience opposite seasons. Earth’s changing distance from the Sun is not the main cause.

By the end of this unit, students should be able to: Students should explain why the two hemispheres have different seasons and reject the distance-from-Sun misconception.

FORMULA AND FACT SUMMARY

Junior quick review

- Earth rotation causes day and night.
- Earth’s axial tilt plus its orbit causes the seasons.
- Moon phases occur because we see different portions of its sunlit half.
- Inverse-square light relationship: intensity is proportional to $1/d^2$; triple the distance gives 1/9 the intensity.
- Four inner planets: Mercury, Venus, Earth, Mars.

Senior quick review

- Kepler third law: $T^2 = a^3$ (T in years; a in AU).
- Parallax distance: d(pc) = 1 / p(arcseconds).
- Flux: F is proportional to L/d^2; compare ratios before substituting values.
- Wien trend: hotter star -> shorter peak wavelength.
- Redshift: z = (observed wavelength - rest wavelength) / rest wavelength.
- Low-redshift estimate: $v \approx cz$; $c \approx 300{,}000$ km/s.

Scope note

These targeted syllabuses are intentionally narrower than full Junior IOAA or International Space Olympiad syllabuses because they are designed to prepare students for the specific practice papers. The organisers’ official examination format and syllabus should be checked separately before representing any material as official.